DEVICE ARCHITECTURE AND WEARABLE CONFIGURATION

General Structural Configuration

The longitudinal penile physiology monitoring device may comprise a ring-form wearable structure configured to maintain positional association with penile anatomy while permitting natural physiologic expansion and contraction. The wearable may be implemented as a fully closed ring, partially open ring, C-shaped configuration, adjustable loop, segmented structure, hinged assembly, elastically deformable band, or other geometry capable of supporting continuous physiologic monitoring without restricting circulation or causing discomfort during extended wear.

In certain embodiments, the device may include an open-gap region between opposing structural arms. The gap may facilitate ease of placement and removal, accommodate anatomical variability, reduce localized pressure concentration, and enable deformation measurement across the gap. The gap region may incorporate magnetic sensing elements, distance sensing structures, compliant mechanical members, or alternative displacement detection mechanisms.

Material and Mechanical Construction

The wearable body may be constructed using rigid materials, flexible polymers, elastomers, silicone-based materials, thermoplastic polyurethane, composite structures, or hybrid layered architectures combining structural support with soft external interfaces. Overmolding techniques may be employed to encapsulate electronics while maintaining user comfort and environmental protection. Mechanical compliance may be distributed circumferentially or localized to specific regions to improve fit stability and long-term wearability.

Embodiments may include interchangeable sizing systems, modular ring diameters, expandable segments, adaptive compression elements, or dynamically compliant structures configured to accommodate anatomical differences among users. The device may be optimized to minimize thickness, weight, and visual profile while maintaining sensing accuracy and mechanical durability.

Electronics Placement and Miniaturization

Electronic components including sensing modules, processing circuitry, wireless communication modules, and power systems may be positioned within a localized housing region or distributed around the wearable structure. Placement may prioritize balance, comfort, and signal integrity. Electronics may be embedded within rigid islands connected by flexible interconnects, flexible printed circuits, or stretchable conductive pathways.

Miniaturization strategies may include integration of multiple sensing functions within shared components, low-power microcontrollers, system-on-chip architectures, stacked circuit assemblies, or custom integrated electronics designed to reduce device volume while preserving functionality.

Power System and Operational Longevity

The device may include a rechargeable or replaceable power source sized to maintain minimal device profile while supporting extended operational periods. Battery placement may occur within a primary housing region, distributed across multiple structural sections, or integrated into flexible portions of the wearable. Charging may occur through electrical contacts, magnetic interfaces, inductive wireless charging, or alternative energy transfer methods.

Operational modes may dynamically adjust sampling rate, sensor activation, and processing frequency to balance measurement fidelity and energy consumption. Modes may include high-fidelity performance operation, balanced monitoring operation, and longevity-focused operation designed to maximize wear duration while preserving longitudinal data continuity.

The architectural embodiments described herein are illustrative and non-limiting. Structural configurations, materials, electronic arrangements, and power architectures may vary without departing from the scope of the invention.